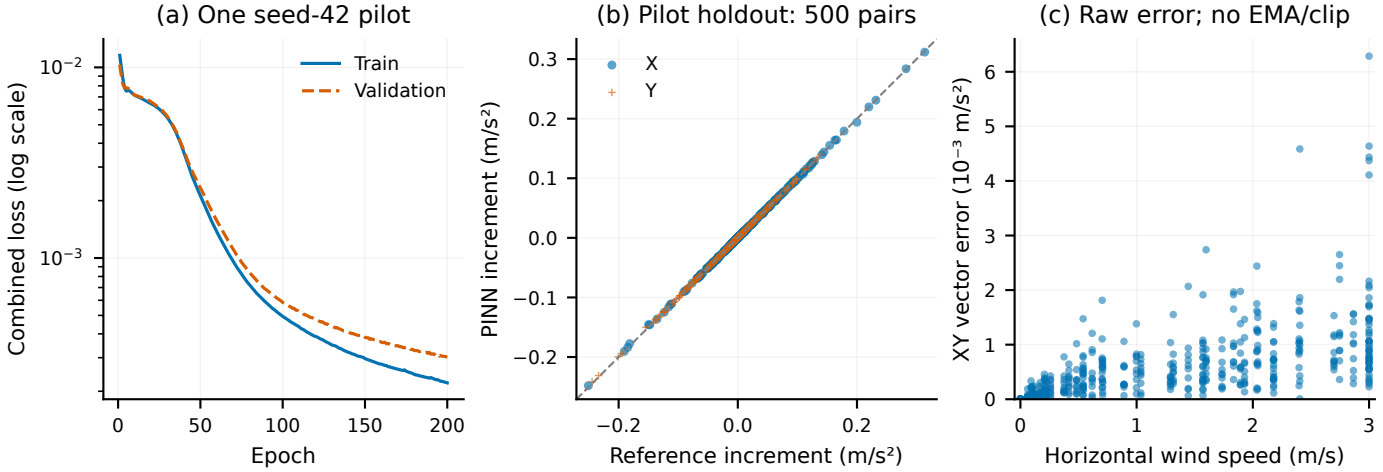
(a) One seed-42 pilot
Train
Validation
Combined loss (log scale)
10^{-2}
10^{-3}
0
50
100
150
200
Epoch
(b) Pilot holdout: 500 pairs
X
Y
PINN increment (m/s²)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.2
0.0
0.2
Reference increment (m/s²)
(c) Raw error; no EMA/clip
XY vector error (10^{-3} m/s²)
6
5
4
3
2
1
0
0
1
2
3
Horizontal wind speed (m/s)